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Question 1 : Write a Program to print no of Capital chars and Small chars in the given String.

CODE →

```
%{      //Declaration Section....
int c = 0, s=0;
%}
%%              //Rule Section
[A-Z] {c++;}
[a-z] {s++;}
. ;
%%
int main(){     //Auxilary Section...
        printf("Enter the String - ");
        yylex();
        printf("Number of Capital chars : %d\n" , c);
        printf("Number of Small chars : %d\n" , s);
        return 0;
}

int yywrap()
{
        return 1;
}
```

06/03/'24

OUTPUT →

```
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ lex capital.l
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ gcc lex.yy.c
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ ./a.out
Enter the String - ALiah UNIversity

Number of Capital chars : 5
Number of Small chars : 10
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$
```

Question 2 : Write a Program to check number of Vowels and Consonants in the given String.

CODE →

```
%{
int v = 0, c=0;
%}
%%
[AEIOUaeiou] {v++;}
[^AEIOUaeiou ] {c++;} //^ not symbol....
%%
int main(){
        printf("Enter the String - ");
        yylex();
        printf("Number of Vowels : %d\n" , v);
        printf("Number of Consonants : %d\n" , c-1);
        return 0;
}

int yywrap()
{
        return 1;
}
```

06/03/'24

OUTPUT →

```
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ lex demo.l
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ gcc lex.yy.c
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ ./a.out
Enter the String - Aliah
Number of Vowels : 3
Number of Consonants : 2
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$
```

Question 3 : Write a Program to print number of Capital Letter Words and Small Letter Words in the given String..

CODE →

```
%{      //Declaration Section....
int c = 0, s=0;
%}
%% //Rule Section
[A-Z]+[^a-z] {c++;}
[a-z]+[^A-Z] {s++;}
.;      // "." means everything .... ";" Eat five star do nothing....
%%
int main(){      //Auxilary Section...
        printf("Enter the String - ");
        yylex();
        printf("Number of Capital Letter Words : %d\n" , c);
        printf("Number of Small Letter Words : %d\n" , s);
        return 0;
}

int yywrap()
{
        return 1;
}
```

06/03/'24

OUTPUT →

```
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ lex capLet.l
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ gcc lex.yy.c
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ ./a.out
Enter the String - ALIAH
Number of Capital Letter Words : 1
Number of Small Letter Words : 0
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$ ./a.out
Enter the String - I aM in ALIAH
Number of Capital Letter Words : 3
Number of Small Letter Words : 1
aliah@aliah-HP-ProDesk-600-G4-SFF:~/LEX$
```

Question 4 : Write a program to print the number of integers and floats

CODE →

```
%{
 int i=0 , f=0;
%}
%%

[0-9]+"."[0-9]+ {f++;}
[0-9]+ {i++;}
. ; //accepts everything
%%
int main(){
printf("Enter numbers: ");
yylex();
printf("Number of integers:%d\n",i);
printf("Number of floats :%d\n",f);
return 0 ;
}
int yywrap(){
return 1 ;
}
```

OUTPUT →

13/03/'24

```
student@student-HP-ProDesk-600-G4-SFF:~$ flex count_int.l
student@student-HP-ProDesk-600-G4-SFF:~$ gcc lex.yy.c
student@student-HP-ProDesk-600-G4-SFF:~$ ./a.out
Enter numbers: 34.5 6 8.9 123 4.77

Number of integers:2
Number of floats :3
student@student-HP-ProDesk-600-G4-SFF:~$
```

Question 5 : Write a program for tokenization of input program.

CODE →

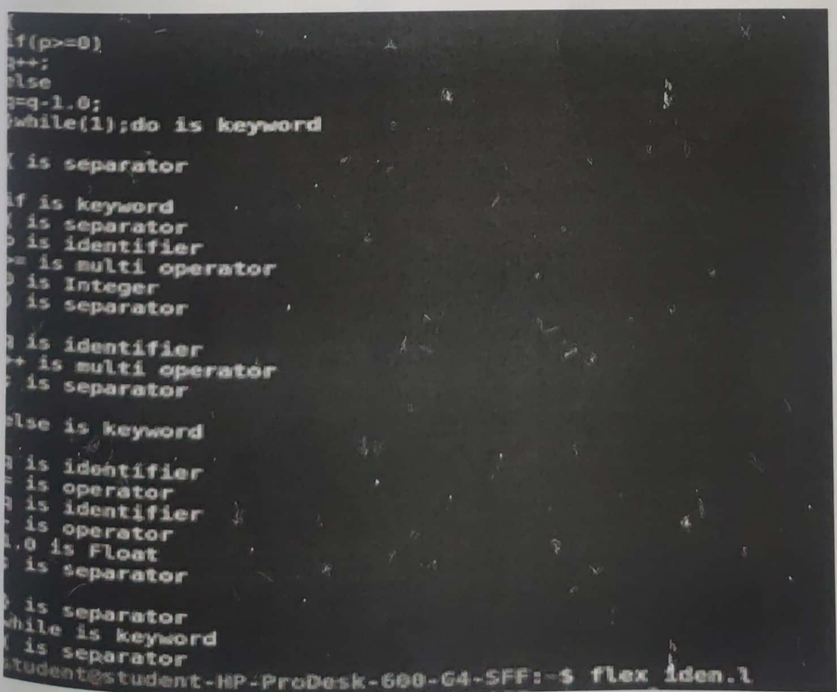
```
%{

%}
%%
[0-9]+"."[0-9]+ {printf("%s is Float \n",yytext);}
[0-9]+ {printf("%s is Integer \n",yytext);}
("do"|"while"|"if"|"else") {printf("%s is keyword \n",yytext);}
[a-zA-Z][a-zA-Z0-9_]* {printf("%s is identifier \n",yytext);}
("++"|">=") {printf("%s is multi operator\n",yytext);}
[=-]+ {printf("%s is operator\n",yytext);}
[(){};] {printf("%s is separator \n",yytext);}
.;
%%
int main(){
printf("Enter strings: ");
yylex();

return 0 ;
}
int yywrap(){
return 1 ;
}
```

13/03/'24

OUTPUT →



```
if(p>=0)
q++;
else
q=q-1.0;
while(1);do is keyword
{ is separator
if is keyword
( is separator
p is identifier
>= is multi operator
0 is Integer
) is separator
q is identifier
++ is multi operator
; is separator
else is keyword
q is identifier
= is operator
q is identifier
- is operator
1.0 is Float
; is separator
} is separator
while is keyword
( is separator
student@student-HP-ProDesk-600-G4-SFF:~$ flex iden.l
```

Question 6 : Write a program for tokenization of input program using file.

CODE →

```
%{

%}
%%
[0-9]+"."[0-9]+ {fprintf(yyout,"%s is Float \n",yytext);}
[0-9]+ {fprintf(yyout,"%s is Integer \n",yytext);}
("do"|"while"|"if"|"else") {fprintf(yyout,"%s is keyword \n",yytext);}
[a-zA-Z][a-zA-Z0-9_]* {fprintf(yyout,"%s is identifier \n",yytext);}
("++"|">=") {fprintf(yyout,"%s is multi operator\n",yytext);}
[=-]+ {fprintf(yyout,"%s is operator\n",yytext);}
[(){};] {fprintf(yyout,"%s is separator \n",yytext);}
.;
%%
int main(){
extern FILE *yyin , *yyout; //input file
yyin=fopen("input","r");
yyout=fopen("output","w");
yylex();
return 0 ;
}
int yywrap(){
return 1 ;
}
```

input file ?

output file ?

~~OUTPUT~~ →

```
{
if(p>=0)
q++;
else
q=q-1.0;
}while(1);do is keyword

{ is separator

if is keyword
( is separator
p is identifier
>= is multi operator
0 is Integer
) is separator

q is identifier
++ is multi operator
; is separator

else is keyword

q is identifier
= is operator
q is identifier
- is operator
1.0 is Float
; is separator

} is separator
while is keyword
( is separator
student@student-HP-ProDesk-600-G4-SFF:~$ flex iden.l
```

13/03/'24

Question 7 : Write a Program to remove all new lines, comment lines, tab and extra spaces from the following piece of code .

CODE →

```
%{
int i;
%}
%%
"//"[^\n]*;
"/*"[^*/]*"*/";
[ \t]* {fprintf(yyout, " ");}
[\n]* {fprintf(yyout, "\n");}
. {fprintf(yyout, "%s", yytext);}
%%
int main(){
        extern FILE *yyin, *yyout;
        yyin = fopen("input1", "r");
        yyout = fopen("output1", "w");
        yylex();
        return 0;
}

int yywrap(){
return 1;
}
```

INPUT-FILE →

input1

```
/*DEFINATION OF CODE*/
int x;
for
do{ //this is comment line
        if (p>=0) q++; //INCREMENT
        else q = q-1.0; //DECREMENT
}while(1); // condition check
```

OUTPUT →

output1

```
int x;
do{
 if (p>=0) q++;
 else q = q-1.0;
}while(1);
```

20/03/'24

Question 8 : Write a Program to accept or reject a String starting with "ab" .

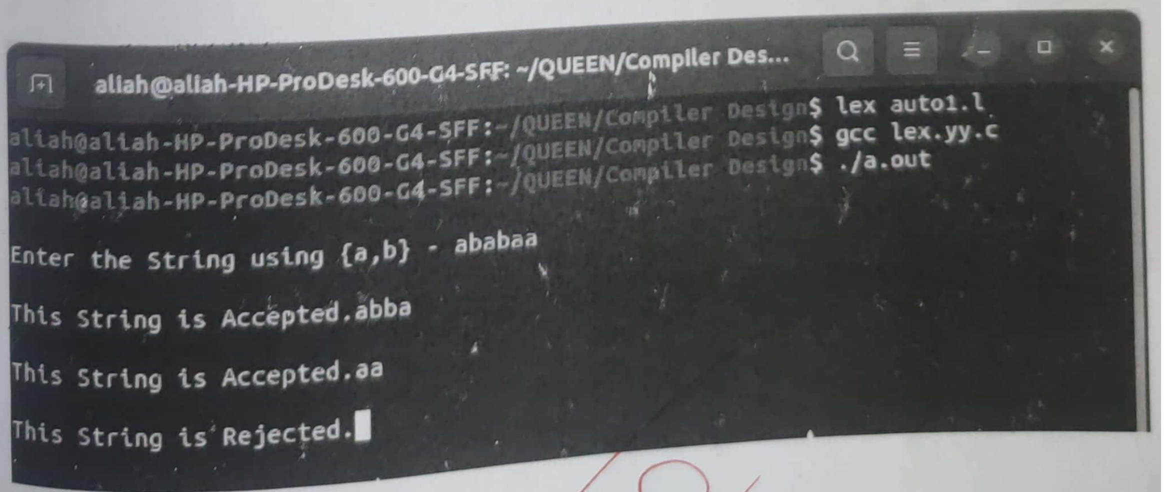
CODE →

```
%s A B C
%%
<INITIAL>a BEGIN A;
<INITIAL>b BEGIN C;
<INITIAL>\n BEGIN INITIAL; printf("\nThis String is Rejected.");
<A>a BEGIN C;
<A>b BEGIN B;
<A>\n BEGIN INITIAL; printf("\nThis String is Rejected.");
<B>a BEGIN B;
<B>b BEGIN B;
<B>\n BEGIN INITIAL; printf("\nThis String is Accepted.");
<C>a BEGIN C;
<C>b BEGIN C;
<C>\n BEGIN INITIAL; printf("\nThis String is Rejected.");
%%

int main(){
printf("\nEnter the String using {a,b} - ");
yylex();
return 0;
}

int yywrap(){
return 1;
}
```

OUTPUT →



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Question 9 : Write a Program to accept or reject a String where "ab" is a Substring .

CODE →

```
%s P Q
%%
<INITIAL>a BEGIN P;
<INITIAL>b BEGIN INITIAL;
<INITIAL>\n BEGIN INITIAL; printf("\nThis String is Rejected.");
<P>a BEGIN P;
<P>b BEGIN Q;
<P>\n BEGIN INITIAL; printf("\nThis String is Rejected.");
<Q>a BEGIN Q;
<Q>b BEGIN Q;
<Q>\n BEGIN INITIAL; printf("\nThis String is Accepted.");
%%

int main(){
printf("\nEnter the String using {a,b} - ");
yylex();
return 0;
}

int yywrap(){
return 1;
}
```

OUTPUT →

```
aliah@aliah-HP-ProDesk-600-G4-SFF:~/QUEEN/Compiler Design$ lex auto2.l
aliah@aliah-HP-ProDesk-600-G4-SFF:~/QUEEN/Compiler Design$ gcc lex.yy.c
aliah@aliah-HP-ProDesk-600-G4-SFF:~/QUEEN/Compiler Design$ ./a.out

Enter the String using {a,b} - abbabbbb

This String is Accepted.aaaaa

This String is Rejected.
```

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Question 10 : Write a program to define integer, float and identifier in NFA.

CODE →

```
%s A B C D
%%
<INITIAL>[0-9] BEGIN A;
<INITIAL>[A-Za-z] BEGIN D;
<INITIAL>\n BEGIN INITIAL; printf("\nInvalid token\n");
<A>[0-9] BEGIN A;
<A>. BEGIN B;
<A>\n BEGIN INITIAL; printf("\nToken-Integer\n");
<B>[0-9] BEGIN C;
<B>\n BEGIN INITIAL; printf("\nInvalid token\n");
<C>[0-9] BEGIN C;
<C>\n BEGIN INITIAL; printf("\nToken-Float\n");
<D>[a-zA-Z0-9_] BEGIN D;
<D>\n BEGIN INITIAL; printf("\nToken-Identifier\n");
%%
int main(){
printf("Enter a string: ");
yylex();
return 0;
}
int yywrap(){
return 1;
}
```

OUTPUT →

```
student1@student-HP-ProDesk-600-G4-SFF:~/Desktop$ ./a.out
enter lexemes - 34234

 Integer constant
1231.342

 Float constant
dsaD34

 Identifiers
34SA
A
 Invalid tokens
student1@student-HP-ProDesk-600-G4-SFF:~/Desktop$
```

Question 11 : Write a yacc code for the given grammar.

CODE →

.l file->

```
%{
#include<stdlib.h>
#include "y.tab.h"
int yylval ;
%}
%%
[0-9]+ {yylval=atoi(yytext); return NUM;}
[+*\n] return *yytext;
. ;
%%
int yywrap()
{
return 1;
}
```

.y file->

```
%{
int yylex(void);
#include<stdio.h>
void yyerror(char *);
%}
%name parse
%token NUM
%left '+'
%left '*' //top to bottom precendence increases
%%
S : S E '\n' {$$=$2;printf("Result is - %d",$$);}
 | {} //epsilon
E : E '+' E {$$=$1+$3;} //':' means equals to
 | E '*' E {$$=$1*$3;}
 |NUM {$$=$1;}
 ;
%%
int main()
{
yyparse();
return 0 ;

oid yyerror(char *msg){
printf("Error in arithmetic");


}
```

OUTPUT →

```
student@student-HP-ProDesk-600-G4-SFF:~$ lex prog1.l
student@student-HP-ProDesk-600-G4-SFF:~$ bison -dy prog1.y
student@student-HP-ProDesk-600-G4-SFF:~$ gcc lex.yy.c y.tab.c
student@student-HP-ProDesk-600-G4-SFF:~$ ./a.out
3+5*1
Result is - 8
```

Question 12 : Write a yacc program to convert infix to postfix expression.

CODE →

.l file->

```
%{
#include<stdlib.h>
#include "y.tab.h"
int yylval ;
%}
%%
[0-9]+ {yylval=atoi(yytext); return NUM;}
[+*\n] return *yytext;
. ;
%%
int yywrap()
{
return 1;
}
```

.y file->

```
%{
int yylex(void);
#include<stdio.h>
void yyerror(char *);
%}
%name parse
%token NUM
%left '+'
%left '*' //top to bottom precendence increases
%%
S : S E '\n' {$$=$2;printf("\n");}
 | {} //epsilon
E : E '+' E {printf("+");} //':' means equals to
 | E '*' E {printf("*");}
 |NUM {printf("%d",$$);}
 ;
%%
int main()
 {
 yyparse();
 return 0 ;
 }
void yyerror(char *msg){
 ...f("Error in arithmetic");
```

OUTPUT →

```
student@student-HP-ProDesk-600-G4-SFF:~$ lex prog1.l
student@student-HP-ProDesk-600-G4-SFF:~$ bison -dy prog1.y
student@student-HP-ProDesk-600-G4-SFF:~$ gcc lex.yy.c y.tab.c
student@student-HP-ProDesk-600-G4-SFF:~$ ./a.out
2+3*5
235*+
```